

```
In [ ]: Dataset name = Country Economy Analysis
```

```
In [5]: import pandas as pd
```

```
In [7]: df = pd.read_excel(r'C:\FSDS_Adv_Gene_AI_&_Agentic_AI\Python\Class_Notes\Pandas_26May\data.xls')
```

```
In [8]: df
```

Out[8]:

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
0	Aruba	ABW	10.244	78.9	High income
1	Afghanistan	AFG	35.253	5.9	Low income
2	Angola	AGO	45.985	19.1	Upper middle income
3	Albania	ALB	12.877	57.2	Upper middle income
4	United Arab Emirates	ARE	11.044	88.0	High income
...	...	...	...	...	...
190	Yemen, Rep.	YEM	32.947	20.0	Lower middle income
191	South Africa	ZAF	20.850	46.5	Upper middle income
192	Congo, Dem. Rep.	COD	42.394	2.2	Low income
193	Zambia	ZMB	40.471	15.4	Lower middle income
194	Zimbabwe	ZWE	35.715	18.5	Low income

195 rows × 5 columns

```
In [9]: id(df) # Memory
```

```
Out[9]: 1345090944176
```

```
In [11]: pd.__version__
```

```
Out[11]: '2.3.3'
```

```
In [12]: len(df)
```

```
Out[12]: 195
```

```
In [18]: df.shape # displays dimension of dataframe
```

```
Out[18]: (195, 5)
```

```
In [16]: df.head() # displays 1st 5 rows
```

Out[16]:

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
0	Aruba	ABW	10.244	78.9	High income
1	Afghanistan	AFG	35.253	5.9	Low income
2	Angola	AGO	45.985	19.1	Upper middle income
3	Albania	ALB	12.877	57.2	Upper middle income
4	United Arab Emirates	ARE	11.044	88.0	High income

```
In [19]: df.head(2) # displays 2 rows
```

```
Out[19]:
```

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
0	Aruba	ABW	10.244	78.9	High income
1	Afghanistan	AFG	35.253	5.9	Low income

```
In [21]: df.tail() # shows bottom 5 rows
```

```
Out[21]:
```

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
190	Yemen, Rep.	YEM	32.947	20.0	Lower middle income
191	South Africa	ZAF	20.850	46.5	Upper middle income
192	Congo, Dem. Rep.	COD	42.394	2.2	Low income
193	Zambia	ZMB	40.471	15.4	Lower middle income
194	Zimbabwe	ZWE	35.715	18.5	Low income

```
In [22]: df.info() # displays information of dataframes
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 195 entries, 0 to 194
Data columns (total 5 columns):
#   Column          Non-Null Count  Dtype
---  -
0   CountryName     195 non-null   object
1   CountryCode     195 non-null   object
2   BirthRate       195 non-null   float64
3   InternetUsers   195 non-null   float64
4   IncomeGroup     195 non-null   object
dtypes: float64(2), object(3)
memory usage: 7.7+ KB
```

```
In [23]: len(df.columns) # number of columns is displayed
```

```
Out[23]: 5
```

```
In [25]: df.dtypes # it displays columnwise datatype of dataframe
```

```
Out[25]: CountryName     object
CountryCode     object
BirthRate       float64
InternetUsers    float64
IncomeGroup     object
dtype: object
```

```
In [27]: df.columns
```

```
Out[27]: Index(['CountryName', 'CountryCode', 'BirthRate', 'InternetUsers',
               'IncomeGroup'],
              dtype='object')
```

```
In [29]: df['CountryName'] # Displays columns data
```

```
Out[29]: 0      Aruba
1      Afghanistan
2      Angola
3      Albania
4      United Arab Emirates
...
190     Yemen, Rep.
191     South Africa
192     Congo, Dem. Rep.
193     Zambia
194     Zimbabwe
Name: CountryName, Length: 195, dtype: object
```

```
In [30]: df['CountryCode']
```

```
Out[30]: 0      ABW
1      AFG
2      AGO
3      ALB
4      ARE
...
190     YEM
191     ZAF
192     COD
193     ZMB
194     ZWE
Name: CountryCode, Length: 195, dtype: object
```

```
In [32]: df[['CountryCode', 'InternetUsers']] # Displays 2 columns details
```

Out[32]:

	CountryCode	InternetUsers
0	ABW	78.9
1	AFG	5.9
2	AGO	19.1
3	ALB	57.2
4	ARE	88.0
...	...	...
190	YEM	20.0
191	ZAF	46.5
192	COD	2.2
193	ZMB	15.4
194	ZWE	18.5

195 rows × 2 columns

```
In [39]: df_numerical_data = df[['BirthRate', 'InternetUsers']]
df_categorcial_data = df[['CountryName', 'CountryCode', 'IncomeGroup']]
```

```
In [40]: df_numerical_data.head(1)
```

Out[40]:

	BirthRate	InternetUsers
0	10.244	78.9

```
In [41]: df_categorcial_data.head(1)
```

Out[41]:

	CountryName	CountryCode	IncomeGroup
0	Aruba	ABW	High income

Slicing in dataframe

```
In [42]: df[:]
```

Out[42]:

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
0	Aruba	ABW	10.244	78.9	High income
1	Afghanistan	AFG	35.253	5.9	Low income
2	Angola	AGO	45.985	19.1	Upper middle income
3	Albania	ALB	12.877	57.2	Upper middle income
4	United Arab Emirates	ARE	11.044	88.0	High income
...	...	...	...	...	...
190	Yemen, Rep.	YEM	32.947	20.0	Lower middle income
191	South Africa	ZAF	20.850	46.5	Upper middle income
192	Congo, Dem. Rep.	COD	42.394	2.2	Low income
193	Zambia	ZMB	40.471	15.4	Lower middle income
194	Zimbabwe	ZWE	35.715	18.5	Low income

195 rows × 5 columns

```
In [43]: df[10:11]
```

Out[43]:

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
10	Azerbaijan	AZE	18.3	58.7	Upper middle income

```
In [44]: df[0:100]
```

Out[44]:

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
0	Aruba	ABW	10.244	78.90	High income
1	Afghanistan	AFG	35.253	5.90	Low income
2	Angola	AGO	45.985	19.10	Upper middle income
3	Albania	ALB	12.877	57.20	Upper middle income
4	United Arab Emirates	ARE	11.044	88.00	High income
...	...	...	...	...	...
95	Korea, Rep.	KOR	8.600	84.77	High income
96	Kuwait	KWT	20.575	75.46	High income
97	Lao PDR	LAO	27.051	12.50	Lower middle income
98	Lebanon	LBN	13.426	70.50	Upper middle income
99	Liberia	LBR	35.521	3.20	Low income

100 rows × 5 columns

In [45]:

df[0:100:20]

Out[45]:

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
0	Aruba	ABW	10.244	78.90	High income
20	Belarus	BLR	12.500	54.17	Upper middle income
40	Costa Rica	CRI	15.022	45.96	Upper middle income
60	Gabon	GAB	30.555	9.20	Upper middle income
80	India	IND	20.291	15.10	Lower middle income

In [46]:

df[:,5]

Out[46]:

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
0	Aruba	ABW	10.244	78.900000	High income
5	Argentina	ARG	17.716	59.900000	High income
10	Azerbaijan	AZE	18.300	58.700000	Upper middle income
15	Bangladesh	BGD	20.142	6.630000	Lower middle income
20	Belarus	BLR	12.500	54.170000	Upper middle income
25	Barbados	BRB	12.188	73.000000	High income
30	Canada	CAN	10.900	85.800000	High income
35	Cameroon	CMR	37.236	6.400000	Lower middle income
40	Costa Rica	CRI	15.022	45.960000	Upper middle income
45	Germany	DEU	8.500	84.170000	High income
50	Ecuador	ECU	21.070	40.353684	Upper middle income
55	Ethiopia	ETH	32.925	1.900000	Low income
60	Gabon	GAB	30.555	9.200000	Upper middle income
65	Gambia, The	GMB	42.525	14.000000	Low income
70	Greenland	GRL	14.500	65.800000	High income
75	Honduras	HND	21.593	17.800000	Lower middle income
80	India	IND	20.291	15.100000	Lower middle income
85	Israel	ISR	21.300	70.800000	High income
90	Kazakhstan	KAZ	22.730	54.000000	Upper middle income
95	Korea, Rep.	KOR	8.600	84.770000	High income
100	Libya	LBY	21.425	16.500000	Upper middle income
105	Lithuania	LTU	10.100	68.452900	High income
110	Moldova	MDA	12.141	45.000000	Lower middle income
115	Mali	MLI	44.138	3.500000	Low income
120	Mozambique	MOZ	39.705	5.400000	Low income
125	Namibia	NAM	29.937	13.900000	Upper middle income
130	Netherlands	NLD	10.200	93.956400	High income
135	Pakistan	PAK	29.582	10.900000	Lower middle income
140	Poland	POL	9.600	62.849200	High income
145	Qatar	QAT	11.940	85.300000	High income
150	Sudan	SDN	33.477	22.700000	Lower middle income
155	El Salvador	SLV	17.476	23.109300	Lower middle income
160	Suriname	SUR	18.455	37.400000	Upper middle income
165	Seychelles	SYC	18.600	50.400000	High income
170	Tajikistan	TJK	30.792	16.000000	Lower middle income
175	Tunisia	TUN	19.800	43.800000	Upper middle income

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
180	Uruguay	URY	14.374	57.690000	High income
185	Virgin Islands (U.S.)	VIR	10.700	45.300000	High income
190	Yemen, Rep.	YEM	32.947	20.000000	Lower middle income

```
In [47]: df[:: -1]
```

Out[47]:

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
194	Zimbabwe	ZWE	35.715	18.5	Low income
193	Zambia	ZMB	40.471	15.4	Lower middle income
192	Congo, Dem. Rep.	COD	42.394	2.2	Low income
191	South Africa	ZAF	20.850	46.5	Upper middle income
190	Yemen, Rep.	YEM	32.947	20.0	Lower middle income
...	...	...	...	...	...
4	United Arab Emirates	ARE	11.044	88.0	High income
3	Albania	ALB	12.877	57.2	Upper middle income
2	Angola	AGO	45.985	19.1	Upper middle income
1	Afghanistan	AFG	35.253	5.9	Low income
0	Aruba	ABW	10.244	78.9	High income

195 rows × 5 columns

```
In [48]: df[:: -5]
```

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
194	Zimbabwe	ZWE	35.715	18.5000	Low income
189	Samoa	WSM	26.172	15.3000	Lower middle income
184	Venezuela, RB	VEN	19.842	54.9000	High income
179	Ukraine	UKR	11.100	41.0000	Lower middle income
174	Trinidad and Tobago	TTO	14.590	63.8000	High income
169	Thailand	THA	11.041	28.9400	Upper middle income
164	Swaziland	SWZ	30.093	24.7000	Lower middle income
159	Sao Tome and Principe	STP	34.537	23.0000	Lower middle income
154	Sierra Leone	SLE	36.729	1.7000	Low income
149	Saudi Arabia	SAU	20.576	60.5000	High income
144	French Polynesia	PYF	16.393	56.8000	High income
139	Papua New Guinea	PNG	28.899	6.5000	Lower middle income
134	Oman	OMN	20.419	66.4500	High income
129	Nicaragua	NIC	20.788	15.5000	Lower middle income
124	Malaysia	MYS	16.805	66.9700	Upper middle income
119	Mongolia	MNG	24.275	20.0000	Upper middle income
114	Macedonia, FYR	MKD	11.222	65.2400	Upper middle income
109	Morocco	MAR	21.023	56.0000	Lower middle income
104	Lesotho	LSO	28.738	5.0000	Lower middle income
99	Liberia	LBR	35.521	3.2000	Low income
94	Kiribati	KIR	29.044	11.5000	Lower middle income
89	Japan	JPN	8.200	89.7100	High income
84	Iceland	ISL	13.400	96.5468	High income
79	Indonesia	IDN	20.297	14.9400	Lower middle income
74	Hong Kong SAR, China	HKG	7.900	74.2000	High income
69	Grenada	GRD	19.334	35.0000	Upper middle income
64	Guinea	GIN	37.337	1.6000	Low income
59	Micronesia, Fed. Sts.	FSM	23.511	27.8000	Lower middle income
54	Estonia	EST	10.300	79.4000	High income
49	Algeria	DZA	24.738	16.5000	Upper middle income
44	Czech Republic	CZE	10.200	74.1104	High income
39	Cabo Verde	CPV	21.625	37.5000	Lower middle income
34	Cote d'Ivoire	CIV	37.320	8.4000	Lower middle income
29	Central African Republic	CAF	34.076	3.5000	Low income
24	Brazil	BRA	14.931	51.0400	Upper middle income
19	Bosnia and Herzegovina	BIH	9.062	57.7900	Upper middle income

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
14	Burkina Faso	BFA	40.551	9.1000	Low income
9	Austria	AUT	9.400	80.6188	High income
4	United Arab Emirates	ARE	11.044	88.0000	High income

Cheking Null Values

In [49]: `df.isnull()`

Out[49]:

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
0	False	False	False	False	False
1	False	False	False	False	False
2	False	False	False	False	False
3	False	False	False	False	False
4	False	False	False	False	False
...	...	...	...	...	...
190	False	False	False	False	False
191	False	False	False	False	False
192	False	False	False	False	False
193	False	False	False	False	False
194	False	False	False	False	False

195 rows × 5 columns

In [50]: `df.isna()`

Out[50]:

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
0	False	False	False	False	False
1	False	False	False	False	False
2	False	False	False	False	False
3	False	False	False	False	False
4	False	False	False	False	False
...	...	...	...	...	...
190	False	False	False	False	False
191	False	False	False	False	False
192	False	False	False	False	False
193	False	False	False	False	False
194	False	False	False	False	False

195 rows × 5 columns

```
In [51]: df.describe() # Displays the statistic values. By default shows on numerical data only.
```

```
Out[51]:
```

	BirthRate	InternetUsers
count	195.000000	195.000000
mean	21.469928	42.076471
std	10.605467	29.030788
min	7.900000	0.900000
25%	12.120500	14.520000
50%	19.680000	41.000000
75%	29.759500	66.225000
max	49.661000	96.546800

```
In [53]: df_categorcial_data.describe()
```

```
Out[53]:
```

	CountryName	CountryCode	IncomeGroup
count	195	195	195
unique	195	195	4
top	Aruba	ABW	High income
freq	1	1	67

Chnaging the name of columns

```
In [60]: df.columns = ['a', 'b', 'c', 'd', 'e']
```

```
In [59]: df.columns
```

```
Out[59]: Index(['CountryName', 'CountryCode', 'BirthRate', 'InternetUsers',  
               'IncomeGroup'],  
              dtype='object')
```

```
In [61]: df.head()
```

```
Out[61]:
```

	a	b	c	d	e
0	Aruba	ABW	10.244	78.9	High income
1	Afghanistan	AFG	35.253	5.9	Low income
2	Angola	AGO	45.985	19.1	Upper middle income
3	Albania	ALB	12.877	57.2	Upper middle income
4	United Arab Emirates	ARE	11.044	88.0	High income

```
In [64]: df.columns = ['CountryName', 'CountryCode', 'BirthRate', 'InternetUsers',  
                      'IncomeGroup']  
df
```

Out[64]:

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
0	Aruba	ABW	10.244	78.9	High income
1	Afghanistan	AFG	35.253	5.9	Low income
2	Angola	AGO	45.985	19.1	Upper middle income
3	Albania	ALB	12.877	57.2	Upper middle income
4	United Arab Emirates	ARE	11.044	88.0	High income
...	...	...	...	...	...
190	Yemen, Rep.	YEM	32.947	20.0	Lower middle income
191	South Africa	ZAF	20.850	46.5	Upper middle income
192	Congo, Dem. Rep.	COD	42.394	2.2	Low income
193	Zambia	ZMB	40.471	15.4	Lower middle income
194	Zimbabwe	ZWE	35.715	18.5	Low income

195 rows × 5 columns

In [62]:

df.columns

Out[62]:

Index(['a', 'b', 'c', 'd', 'e'], dtype='object')

In [65]:

df.head()

Out[65]:

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
0	Aruba	ABW	10.244	78.9	High income
1	Afghanistan	AFG	35.253	5.9	Low income
2	Angola	AGO	45.985	19.1	Upper middle income
3	Albania	ALB	12.877	57.2	Upper middle income
4	United Arab Emirates	ARE	11.044	88.0	High income

Multiplying 2 columns

In [74]:

df['My cal'] = df.BirthRate * df.InternetUsers

In [82]:

df.head()

Out[82]:

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup	My cal	My cal
0	Aruba	ABW	10.244	78.9	High income	808.2516	808.2516
1	Afghanistan	AFG	35.253	5.9	Low income	207.9927	207.9927
2	Angola	AGO	45.985	19.1	Upper middle income	878.3135	878.3135
3	Albania	ALB	12.877	57.2	Upper middle income	736.5644	736.5644
4	United Arab Emirates	ARE	11.044	88.0	High income	971.8720	971.8720

Dropping of column

```
In [91]: df = df.drop('My cal', axis= 1)
df
```

```
Out[91]:
```

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup	Mycal
0	Aruba	ABW	10.244	78.9	High income	808.2516
1	Afghanistan	AFG	35.253	5.9	Low income	207.9927
2	Angola	AGO	45.985	19.1	Upper middle income	878.3135
3	Albania	ALB	12.877	57.2	Upper middle income	736.5644
4	United Arab Emirates	ARE	11.044	88.0	High income	971.8720
...	...	...	...	...	...	...
190	Yemen, Rep.	YEM	32.947	20.0	Lower middle income	658.9400
191	South Africa	ZAF	20.850	46.5	Upper middle income	969.5250
192	Congo, Dem. Rep.	COD	42.394	2.2	Low income	93.2668
193	Zambia	ZMB	40.471	15.4	Lower middle income	623.2534
194	Zimbabwe	ZWE	35.715	18.5	Low income	660.7275

195 rows × 6 columns

```
In [95]: df = df.drop('Mycal', axis = 1)
df
```

```
Out[95]:
```

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
0	Aruba	ABW	10.244	78.9	High income
1	Afghanistan	AFG	35.253	5.9	Low income
2	Angola	AGO	45.985	19.1	Upper middle income
3	Albania	ALB	12.877	57.2	Upper middle income
4	United Arab Emirates	ARE	11.044	88.0	High income
...	...	...	...	...	...
190	Yemen, Rep.	YEM	32.947	20.0	Lower middle income
191	South Africa	ZAF	20.850	46.5	Upper middle income
192	Congo, Dem. Rep.	COD	42.394	2.2	Low income
193	Zambia	ZMB	40.471	15.4	Lower middle income
194	Zimbabwe	ZWE	35.715	18.5	Low income

195 rows × 5 columns

Filtering in dataframes

```
In [102... df[df['InternetUsers'] < 2 ]
```

Out[102...

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
11	Burundi	BDI	44.151	1.3	Low income
52	Eritrea	ERI	34.800	0.9	Low income
55	Ethiopia	ETH	32.925	1.9	Low income
64	Guinea	GIN	37.337	1.6	Low income
117	Myanmar	MMR	18.119	1.6	Lower middle income
127	Niger	NER	49.661	1.7	Low income
154	Sierra Leone	SLE	36.729	1.7	Low income
156	Somalia	SOM	43.891	1.5	Low income
172	Timor-Leste	TLS	35.755	1.1	Lower middle income

In [103...

```
len(df[df['InternetUsers'] < 2 ])
```

Out[103...

9

In [104...

```
df[df['BirthRate'] > 40 ]
```

Out[104...

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
2	Angola	AGO	45.985	19.1	Upper middle income
11	Burundi	BDI	44.151	1.3	Low income
14	Burkina Faso	BFA	40.551	9.1	Low income
65	Gambia, The	GMB	42.525	14.0	Low income
115	Mali	MLI	44.138	3.5	Low income
127	Niger	NER	49.661	1.7	Low income
128	Nigeria	NGA	40.045	38.0	Lower middle income
156	Somalia	SOM	43.891	1.5	Low income
167	Chad	TCD	45.745	2.3	Low income
178	Uganda	UGA	43.474	16.2	Low income
192	Congo, Dem. Rep.	COD	42.394	2.2	Low income
193	Zambia	ZMB	40.471	15.4	Lower middle income

In [105...

```
len(df[df['BirthRate'] > 40 ])
```

Out[105...

12

In []:

```
# Merging in dataframe
```

In [117...

```
df[(df.BirthRate > 40 ) & (df.InternetUsers < 2)]
```

Out[117...

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
11	Burundi	BDI	44.151	1.3	Low income
127	Niger	NER	49.661	1.7	Low income
156	Somalia	SOM	43.891	1.5	Low income

In []:

In []: